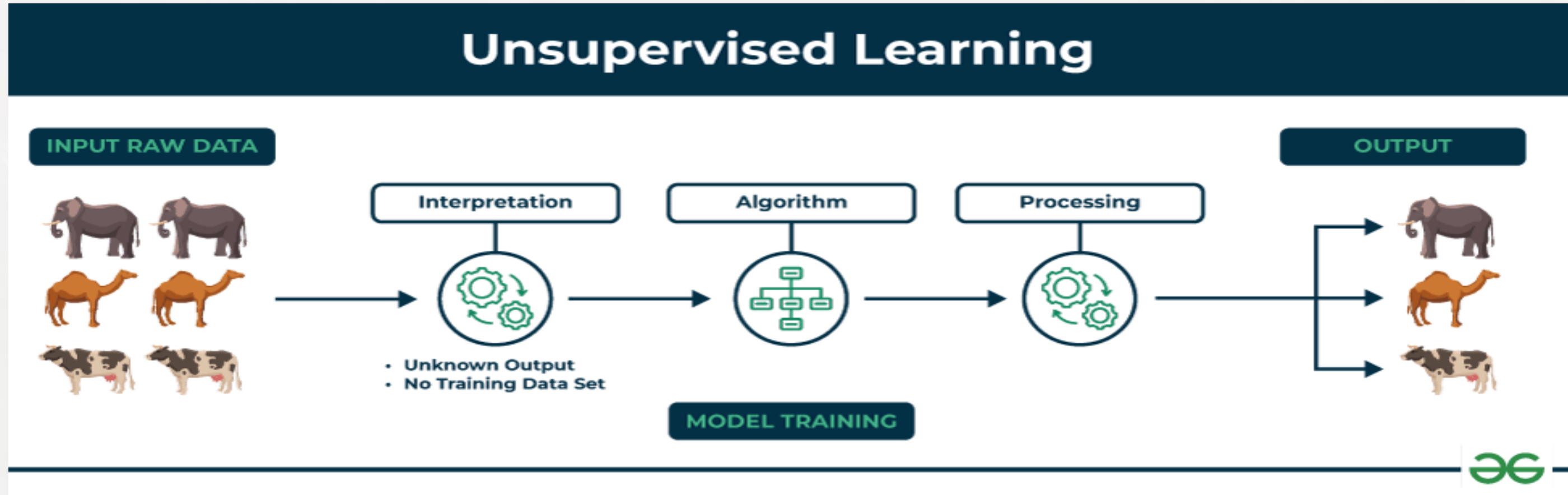


Unsupervised Learning

KMEANS CLUSTERING ALGORITHM
DBSCAN CLUSTERING ALGORITHM

Anuraj Sapkota

Unsupervised Learning



- Training models on unlabeled data to find hidden patterns and relationships.
- Don't know output variable.
- Similar kind of points will be grouped together and form a cluster

-
- Used for: Customer Segmentation, Anomaly detection, Image compression.
 - It explores data intrinsically

Advantages

- No need of Labeled data.
- Discover hidden pattern
- Detects outliers in data that don't fit existing clusters
- Used for exploring new dataset.

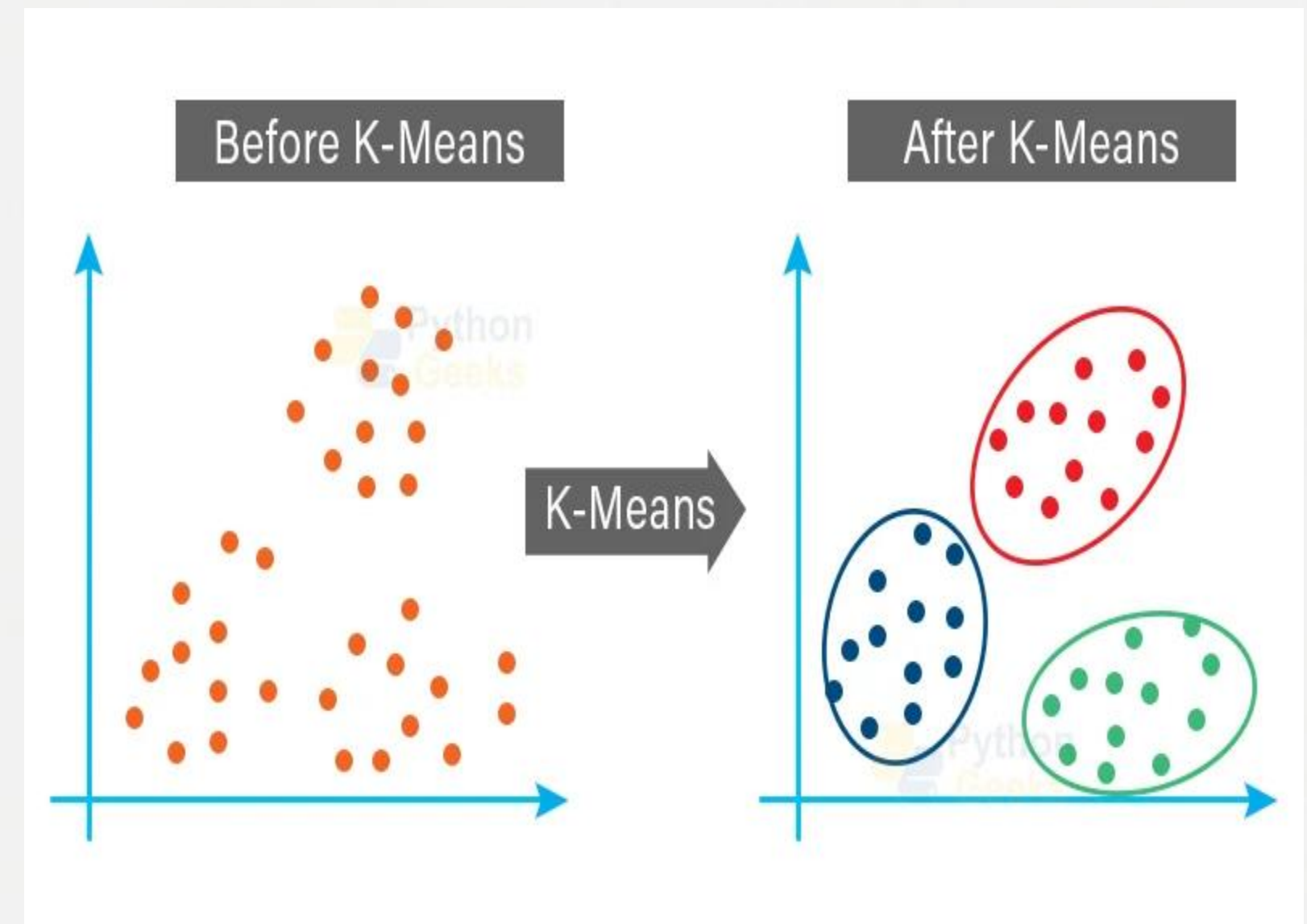
K-means Clustering

K indicates the number of predefined clusters that need to create.

Centroid based algorithm.

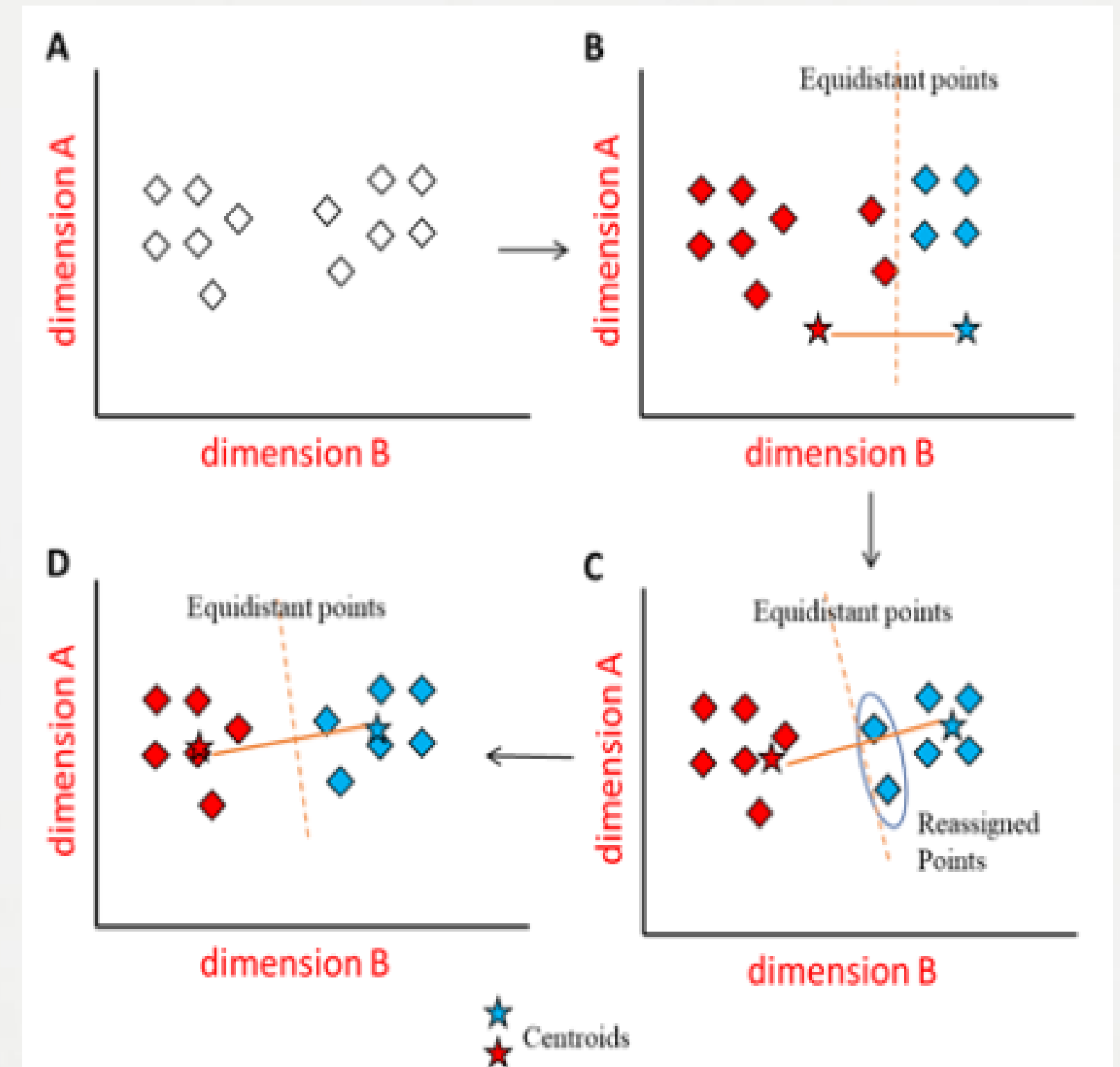
Aims to minimize the sum of distance between datapoints and their corresponding clusters.

Group data into K clusters by minimizing Intra-cluster variance



How it works ?

- Initialize some k (Centroids).
- Using Euclidean distance, points that are nearest to the centroid are grouped
- Recalculate centroids as mean of points in each cluster.
- Repeat until centroid moves at center of group or centroids stabilize.
- The algorithm minimizes the sum of squared distance (inertia)



Parameters

- K: Numer of clusters. Must be chosen in advanced. Elbow method or silhouette score
- Distance metric: Euclidean by default.

Advantage

- It is very straight forward and easy to understand and implement.
- Scalable to large dataset
- Easy to interpret result.

Disadvantage

- Sensitive in assigning initial centroids.
- Assuming spherical clusters fails on irregular shapes.
- Outlier can skew results

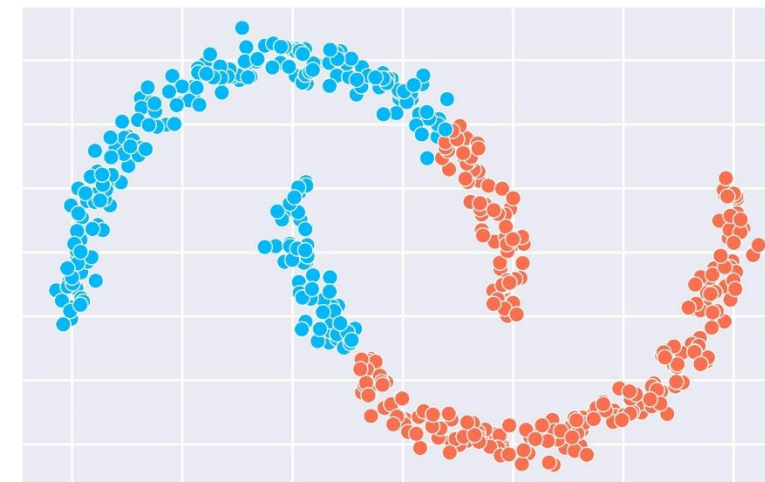
Random Initialization Trap

- Occurs when starting centroids are chosen badly.
- Leads to converge at local minima.
- Centroid very near or very far to each other also lead to this trap.
- Kmeans++ is used to mitigate this problem.
- Kmeans++ assigns on the basis of probability proportional to sq. distance from nearest existing centers.

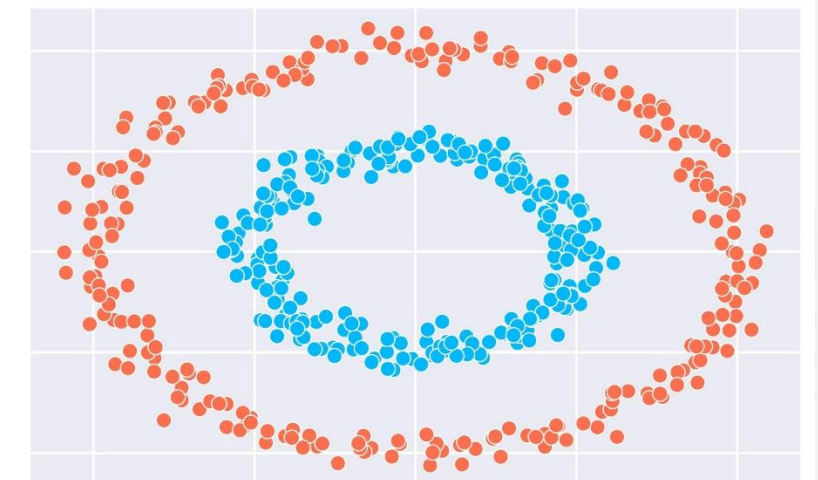
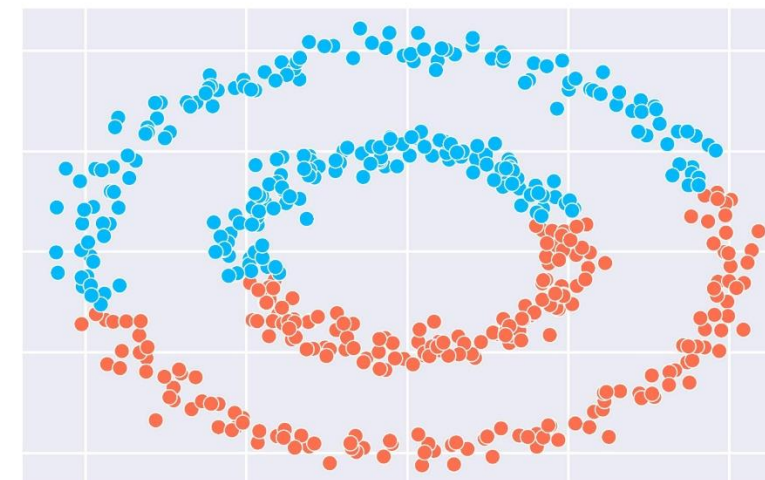
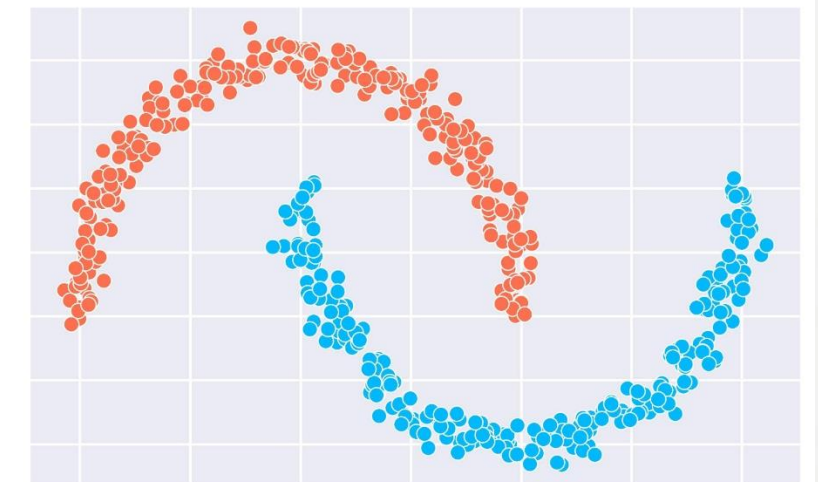
DBSCAN Clustering

- Density-Based Spatial Clustering of Application of Noise
- A density based clustering algorithm that group points in high-density areas.
- Detect arbitrary clusters.
- It is robust to outliers.

KMeans

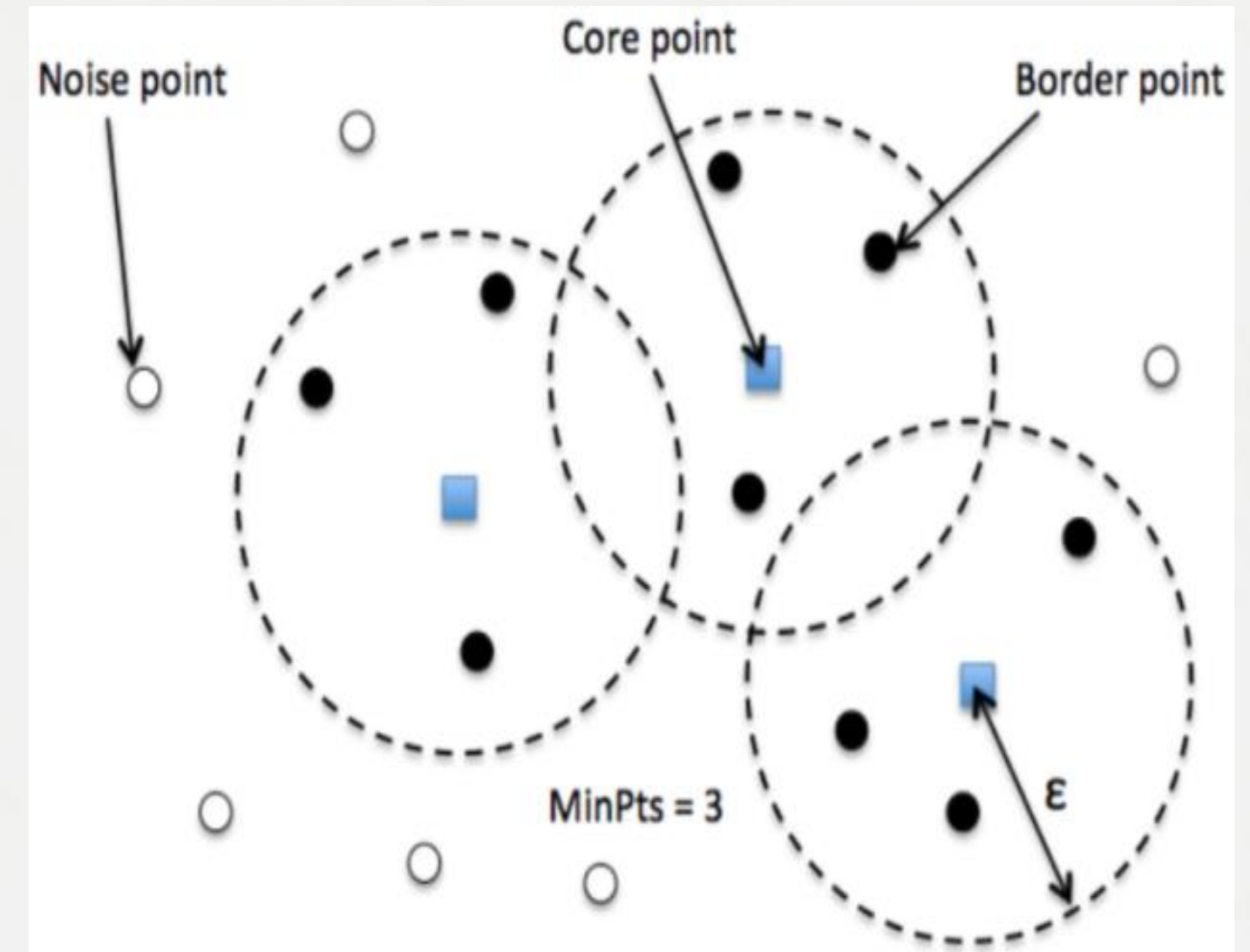


DBSCAN



How it works ?

- **Core Point:** Point having neighbors equal to or greater than Minimum_pts inside epsilon or radius.
- **Border Point:** Point having neighbors less than minimum_pts.
- **Noise:** Point: Neither a core point nor a border point



Steps

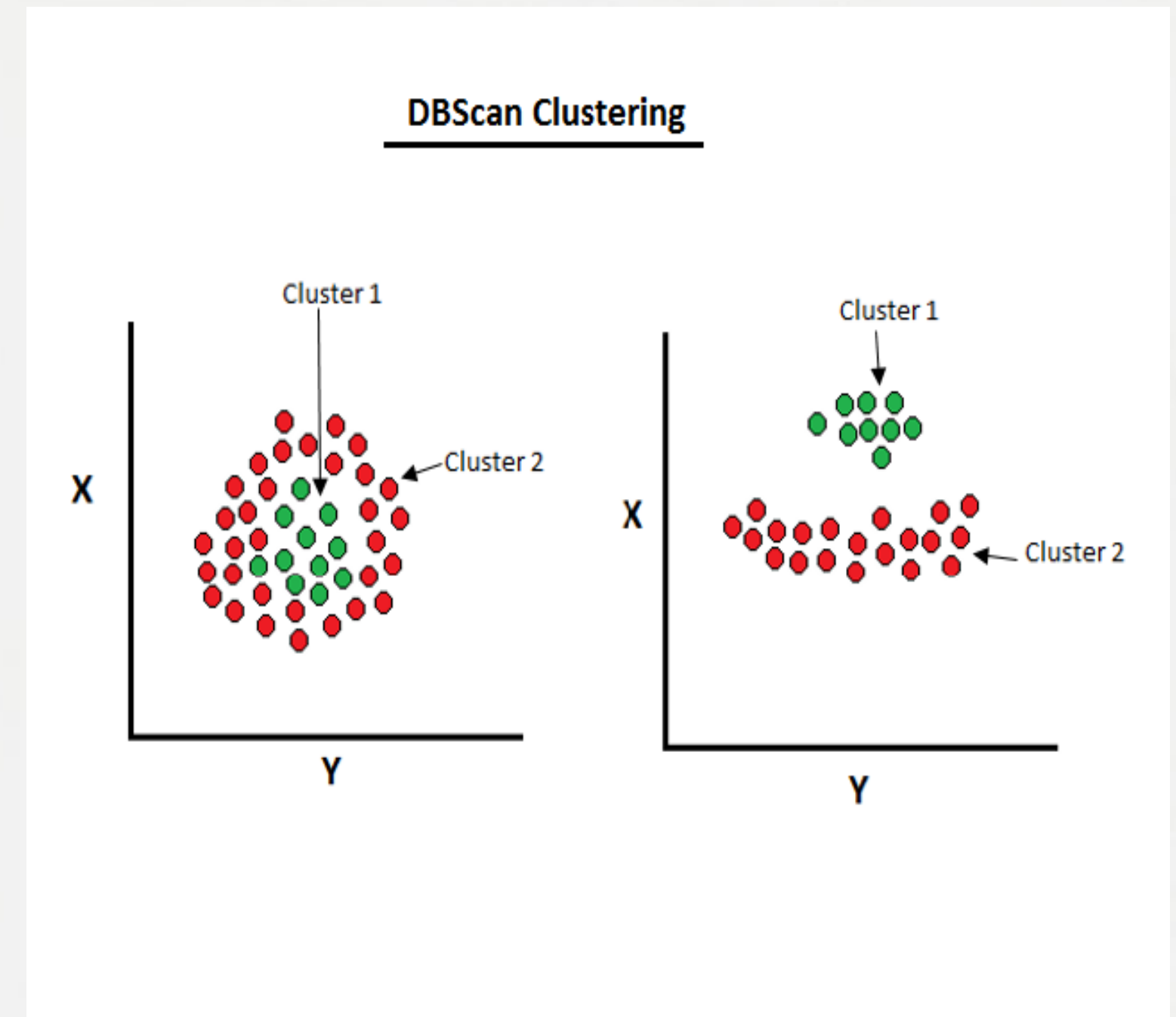
- Define `eps` and `min_pts`.
- Identify core points and form a new cluster.
- For each core point, expand the cluster by visiting neighboring points within `eps`.
- Classify border point cluster doesn't go beyond border point.
- Continue until all points are visited and assigned to a cluster or marked as noise.

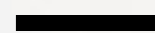
Parameters

- Eps: Maximum distance between cluster.
- MinPts: Minimum Points to form a dense region.

Advantages

- Handles arbitrary shaped clusters.
- Robust to outliers.
- No need to specify no of clusters.





Disadvantages

- Struggles with datasets of varying densities.
- Higher computational complexity.
- Not useful for high dimensional data without preprocessing.

The End